Passive ISAC for Urban Transit using Self-Supervised World Models

The inMotion Project

The inMotion Project

The **inMotion** project aims to bring real-time intelligence to public transport operations. One of its core goals, understand **how passengers move** through the network: who boards, who alights, and where.
A previous paper answered that with hand-crafted RSSI features and supervised classifiers. Today the question is different: can a **world model learn the dynamics** itself?



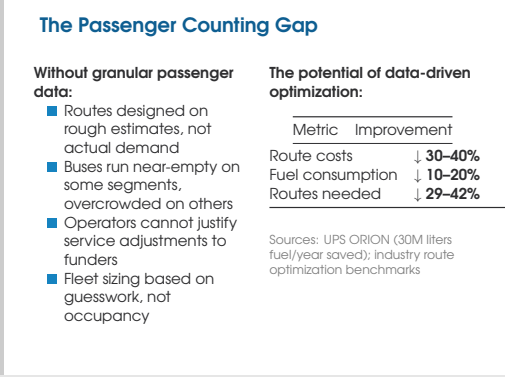
Smart bus concept

- Good morning. This work is part of the inMotion project, supported by CENTRO 2030.
- The project's mission is to give public transport operators real-time knowledge of how passengers move: who boards, who alights, and where.
- At EuCNC we showed that this is possible with hand-crafted RSSI features and supervised classifiers. The ceiling was a Matthews Correlation Coefficient of 0.756.
- This talk is about what changed: instead of engineering the features, we let a self-supervised world model learn the radio dynamics, and the metric goes from 0.756 to 0.9194.
- Everything you see is one access point already on the bus. No camera, no radar, no new hardware.

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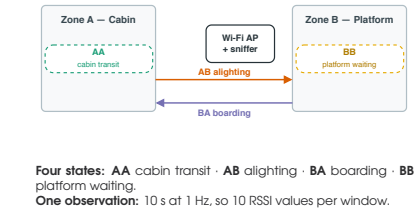
└─The Passenger Counting Gap



- Accurate passenger counting is the input to every operational decision an operator makes.
- Traditional systems, infrared beams, pressure mats, cameras, work reasonably in the lab. In the field, with crowds, occlusion and dirt, claimed 98 percent becomes 53 percent. Pronello and Ruiz documented that gap.
- But the financial case is what convinces operators. UPS reports that ORION cuts about 100 million miles and 10 million gallons of fuel per year. In public transport the same exercise is documented: a linear-programming redesign of the Queretaro bus network removed 24 of its 148 routes and saved almost 14,000 litres of diesel, 37 tonnes of CO2 and 17,000 dollars per day.
- These are measured outcomes, not projections. If you know exactly where people board and alight, you can right-size vehicles and cut empty runs. The numbers on the right are

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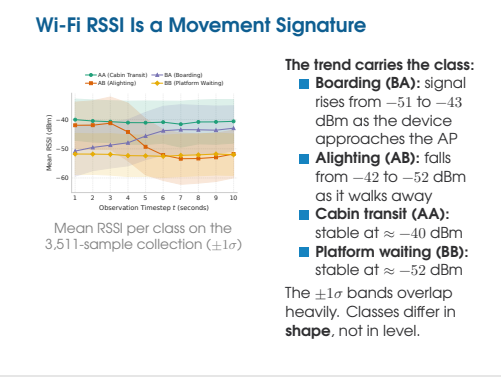
Four Movements to Classify



- Formalise the problem. There is a door, and one access point sits right at it.
- Zone A is the cabin: an enclosed room, strong signal, roughly minus 40 dBm. Zone B is the platform: a corridor behind a wall, roughly minus 52 dBm.
- Every person near the door is in one of four states. Inside and staying inside is AA, cabin transit. Platform and staying is BB, platform waiting. Moving from the platform into the cabin is BA, boarding. Moving out is AB, alighting.
- The physics is in the level, but the semantics is in the *trend*. That is the whole difficulty.
- One observation window is ten seconds sampled at one hertz, so ten RSSI values. That is the raw input to everything I show later.

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└Wi-Fi RSSI Is a Movement Signature



- This is why the problem is solvable at all, and also why it is hard.
- Boarding, the purple line, climbs about eight dBm as the person walks toward the access point. Alighting, the orange line, falls about ten dBm as they walk away.
- The two static classes sit at different levels: AA is high because the device is inside near the AP, BB is low because it is outside behind a wall.
- Now look at the shaded bands. The standard deviations overlap heavily. If you only looked at the level, you could not separate boarding from standing inside, because both are strong.
- What separates them is the temporal shape. That is exactly what a world model is good at learning, and exactly what a hand-built threshold is bad at.

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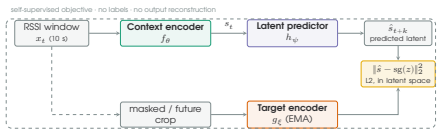
How a World Model Learns



- This is the slide to remember, and there is no maths on it.
- Forget networks for a second. Think of a child learning what a ball does. Nobody hands it a labelled dataset. It watches, guesses what happens next, and is surprised when it is wrong. That is all a world model is, formalised.
- Here the child sees ten seconds of radio. We hide the second half and make it guess. It is scored only on that guess.
- The non-obvious part: it guesses in its own compressed internal language, not in the signal. We do not ask it to redraw the waveform. Redrawing the waveform means redrawing the noise, and the noise is exactly what differs between one bus, one phone and one season.
- The crazy part: nobody ever tells it that this window is a boarding. It never sees the word. And that is enough: it beats every supervised model we trained on this task, and it

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Predicting in Latent Space

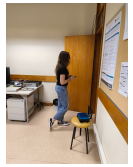


- Now the machinery, for the record.
- The left branch encodes a window into a latent vector. The predictor is asked for the latent of a masked or future piece of the same window.
- The bottom branch is the same encoder, kept as a slowly updated EMA copy, and its output goes through a stop-gradient. That asymmetry is what stops the two branches from collapsing onto the same trivial answer.
- The loss on the right is an L2 distance between two latent vectors. No label, no reconstruction, no decode.
- Why this matters: the prediction problem forces the representation to keep what physically evolves and to drop what does not.
- The four variants in the paper differ only in how they tokenise time and where they put the mask. The idea is identical.

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Data Collection Setup

Data Collection Setup



Controlled indoor environment, IT Aveiro

Physical setup:
- Zone A: closed room (cabin), AP at the doorway
- Zone B: corridor (platform), wall attenuation between zones

Collection:
- 10 s windows at 1 Hz → 10 RSSI values
- **7 smartphone models**, multiple vendors, multiple Android releases
- Scripted movements with ground-truth session labels
- Passive sniffing: no association, no identification

- We built a controlled environment at the institute. A closed room represents the cabin, and the corridor outside represents the platform.
- The access point sits exactly at the doorway, where it would be mounted on a vehicle.
- We collected from seven distinct smartphone models across multiple vendors and several Android releases. This matters: the radio chipset changes the RSSI curve, and a model that has only seen one phone will not generalise.
- Each trial is ten seconds, sampled once per second.
- Movements were scripted and the label comes from the session protocol, not from a second annotator. Keep that in mind when we discuss limitations.
- The sniffer is passive. We never associate with the phone, we never ask it to identify itself.

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└Dataset Composition and Splits

Dataset Composition and Splits

Condition	Samples	Description
Single-device	160	pure signals, 40 per class
Multi-device	3,351	concurrent interference, ≈ 840 per class
Total	**3,511**	primary benchmark

Strict superset. The earlier 2,251-sample collection is fully contained: all eight earlier devices are retained, and five more are added.

Protocol. 80/20 class-stratified split, with 12.5% of training held out for validation. Gaussian noise ($\sigma = 0.05$) and intra-class Mixup ($\alpha = 0.4$) give **35,839** training windows; the test split is evaluated once.

- The primary dataset is 3,511 labelled windows. 160 are single-device calibration traces with clean signals, 40 per class. The other 3,351 are multi-device traces, roughly 840 per class, where up to four phones transmit at the same time.
- The noisy part is the realistic part. On a real platform you have people boarding, waiting and walking past, all with active phones.
- This collection is a strict superset of the 2,251-sample campaign from our earlier paper: every device is retained and five more are added. That is deliberate. It lets us measure what happens to each model family when the collection grows.
- Augmentation happens inside the training folds only, with Gaussian noise and Mixup. That gives about 36,000 training windows with no contamination of the split.

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└The Four World Models

- All four models share one objective: predict the representation of a held-out region of the window, in latent space, without labels.
- TS-JEPA is our strongest variant. It uses 18 input channels, a 1D convolutional patch tokenizer with patch size two, and patch masking annealed from 0.4 to 0.7 during pretraining.
- LeJEPA is the compact one. Four base channels, causal next-step prediction, no EMA target: it is stabilised instead by a sketched isotropic Gaussian regulariser. It has 0.566 million parameters, which is why it is the edge candidate.
- CF-JEPA predicts across three horizons without any masking, and T-JEPA tokenises each feature separately with a learned register token.
- The baselines matter for the story: the strongest supervised and state-space models, tuned with Optuna and given equal compute, are what we compare against.

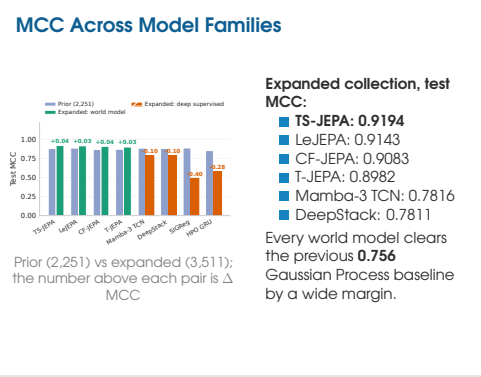
The Four World Models

	TS-JEPA	LeJEPA	CF-JEPA
Input channels	18 (rich)	4 (base)	18 (rich)
Tokenizer	1D conv patch	causal temporal	1D conv patch
Patch / token size	2 timesteps	1 timestep	2 timesteps
Embedding dim / heads	256 / 8	128 / 4	256 / 8
Encoder layers	4	4	3
Predictive objective	masked patch L2	next-step L2	multi-horizon L2
Anti-collapse	EMA target	SIGReg	EMA target
Parameters	2.153 M	**0.566 M**	1.232 M

Baselines: Mamba-3 state spaces, four front-ends (0.84–1.70 M) · **SIGReg:** directly supervised CNN with the SIGReg regulariser, *no pretraining* (0.697 M) · DeepStack (43.6 M) · Optuna-tuned GRU.
Interpretability: SHAP attributions over the 18 channels, plus architecture-controlled ablation.

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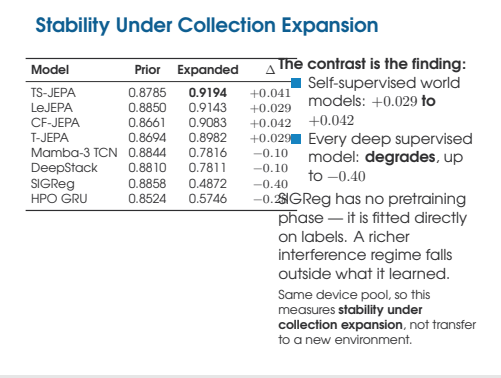
└MCC Across Model Families



- Here is the headline. On the expanded collection the four world models take the top four places, from 0.9194 down to 0.8982 in MCC.
- The previous best result on this task was 0.756, from a Gaussian Process on hand-crafted features. We are not comparing against a weak baseline: the Gaussian Process was the winner of a 38-classifier study.
- The strongest supervised deep models, Mamba-3 TCN and DeepStack, land around 0.78. So the gap is not small, and it is not noise: three seeds, standard deviations of about 0.008.
- Green bars are the world models, orange are the deep supervised and state-space models. The hatched bars are the ones that degrade as the collection grows, which is the next slide.
- Note that the bars are on the log-ish scale of MCC, so a 0.14 gap is a large change in error structure, not a rounding

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└Stability Under Collection Expansion



- This table is the scientific core of the paper, more than the headline number.
- Look at the direction of the arrows. All four self-supervised models *improve* when the collection grows, by three to four hundredths of MCC. Every deep supervised and state-space model *degrades*, and SIGReg collapses from 0.8858 to 0.4872.
- Note first what SIGReg is: a directly supervised CNN with the SIGReg latent regulariser and no pretraining phase. It is in our model registry because it shares the regulariser, not because it is self-supervised.
- Why does it break? Its regulariser constrains the latent space around the training distribution, and the boundaries are fitted to artefacts of the old training set. When the interference regime becomes richer, they break.
- The world models never optimise a label-conditioned

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Stacking and Calibration

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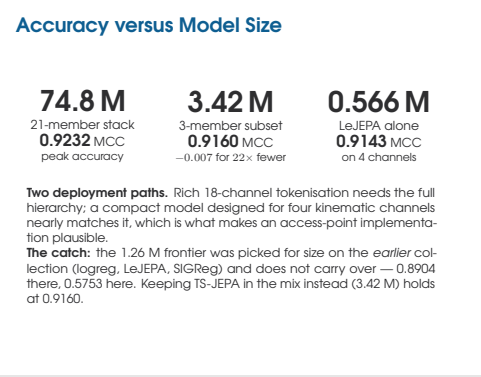
Ensemble	n	Params	Prior	Expanded	Noise /
21-member full stack	21	74.8 M	0.8873	**0.9232**	**0.9563** /
16-member subset	16	73.2 M	0.8873	**0.9232**	**0.9563** /
World-only stack	7	6.68 M	0.8872	0.9065	0.9450 /
Best-world subset	3	3.42 M	0.8784	0.9160	0.9341 /
Three-member Pareto	3	**1.26 M**	**0.8904**	0.5753	0.9396 /

- Every member is **temperature-calibrated** on validation NLL, then fused by an L2 multinomial logistic-regression meta-learner.
- The 16-member non-redundant subset matches the full 21-member stack exactly: the weak recurrent backbones add nothing.
- Ensembles are pruned by a greedy **Pareto search**: MCC against parameter count.

- Stacking the models does help, but the interesting part is where the gain stops.
- Every member is temperature-calibrated first: one scalar temperature per model, fitted on validation negative log-likelihood. This matters because uncalibrated softmax outputs are overconfident exactly under the interference we care about.
- The 21-member stack reaches 0.9232 on the expanded collection. On the multi-device subset alone it reaches 0.9563, and on pure single-device traces 0.9596.
- The 16-member non-redundant subset matches the full stack exactly: 0.9232. So the weak recurrent backbones contribute nothing.
- Now the catch in the small print. The size-optimal frontier from the earlier collection was logreg plus LeJEPA plus SIGReg: 1.26 million parameters, 0.8904 there. Carried to the

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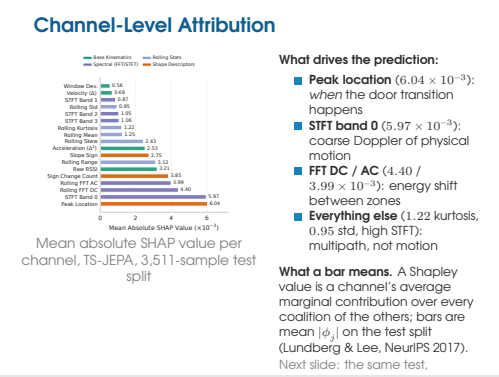
└Accuracy versus Model Size



- Let me put the parameter question in perspective.
- The full stack needs 74.8 million parameters and gives 0.9232.
- A three-member subset — SIGReg, LeJEPA and TS-JEPA — needs 3.42 million and gives 0.9160. That is seven thousandths of MCC for twenty-two times fewer parameters.
- And LeJEPA alone, with 0.566 million parameters over only four channels, gives 0.9143. Essentially the same accuracy as the ensemble, at a hundred and thirty times fewer parameters than the stack.
- That is not a small engineering detail: it is the difference between a datacentre model and something that runs on the access point you already have on the bus.
- The trap on the bottom is worth remembering: optimising for size on one collection does not survive a change in regime, because the Pareto frontier inherited a model that collapses.

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└Channel-Level Attribution

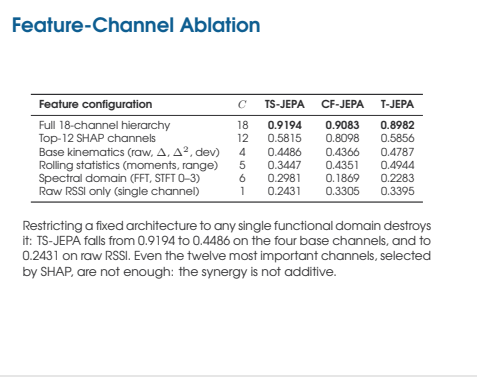


Mean absolute SHAP value per channel, TS-JEPA, 3,511-sample test split

- Shapley Additive exPlanations, from cooperative game theory: Lundberg and Lee, NeurIPS 2017. A Shapley value is a player's average marginal contribution over every possible coalition of the other players; here the players are the eighteen channels, and the bars are the mean absolute value over the test split.
- The decomposition is exact: the eighteen attributions add up to the model output minus its expected value, so the bars account for the whole prediction, not a subset of it.
- Attributions are computed on the leading TS-JEPA model over the 18-channel hierarchy.
- Two features dominate: the peak location index, which is essentially the timestamp of the door transition, and the lowest STFT band, which captures the coarse Doppler shift of a body in motion.
- Then come the DC and AC energies of the rolling FFT: those

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└Feature-Channel Ablation



Feature-Channel Ablation

Feature configuration	C	TS-JEPA	CF-JEPA	T-JEPA
Full 18-channel hierarchy	18	**0.9194**	**0.9083**	**0.8982**
Top-12 SHAP channels	12	0.5815	0.8098	0.5856
Base kinematics (raw, Δ, Δ^2, dev)	4	0.4486	0.4366	0.4787
Rolling statistics (moments, range)	5	0.3447	0.4351	0.4944
Spectral domain (FFT, STFT 0–3)	6	0.2981	0.1869	0.2283
Raw RSSI only (single channel)	1	0.2431	0.3305	0.3395

Restricting a fixed architecture to any single functional domain destroys it: TS-JEPA falls from 0.9194 to 0.4486 on the four base channels, and to 0.2431 on raw RSSI. Even the twelve most important channels, selected by SHAP, are not enough: the synergy is not additive.

- This is the ablation that surprised us, and it is the strongest evidence for the world-model framing.
- If you keep the architecture identical and only change the input, removing channels is catastrophic. TS-JEPA goes from 0.9194 with eighteen channels, to 0.4486 with the four base kinematic channels, to 0.2431 with raw RSSI alone.
- The interesting row is the second: the top twelve channels ranked by SHAP give 0.5815. So it is not that twelve channels are missing some information; it is that a patch tokenizer over a rich set learns cross-channel interactions that a reduced set cannot express.
- Consequence for deployment: you cannot simply strip a rich model down. Either you keep the hierarchy, or you use an architecture designed for a small input, like LeJEPA, which gets 0.9143 on four channels with 0.566 million parameters.
- In other words, the two deployment paths are real, and the

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Privacy Through Edge Inference

Privacy Through Edge Inference

The threat is real:

- Passive sniffing needs no cooperation and no visible trace
- Radio sequences survive MAC randomisation
- Commuter regularity links a passenger across days
- Profiling defeats naive anonymisation and DP (Silva et al.)

What the architecture buys:

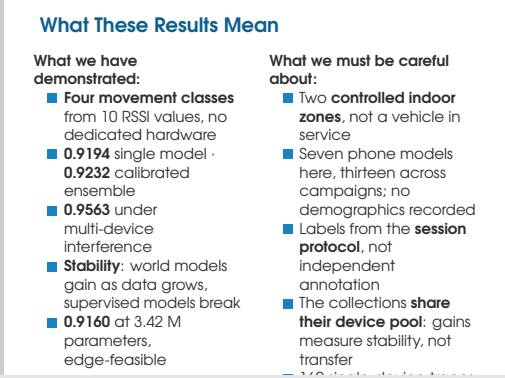
- 0.9143 MCC at **0.566 M** parameters: small enough for AP firmware
- Design target: infer *inside* the AP (deployment is future work)
- Raw RSSI and latents live in RAM and are discarded
- Only aggregated counts leave: Δboarding, Δalighting

The accuracy that makes the sensing useful is the same property that makes edge confinement plausible.

- Passive sensing avoids cameras and microphones, but it is not automatically private. Any receiver in range can capture the same frames we do, and it can do so without the operator knowing.
- The literature is clear that anonymisation is not enough. Silva and colleagues showed that profiling models recover behavioural patterns from interaction data, and that differential privacy can be defeated when a classifier exploits temporal correlations. In transit, the regular commute is exactly such a correlation: the same person, the same door, the same time of day.
- Our mitigation is architectural rather than legal. LeJEPA reaches 0.9143 with 0.566 million parameters, which is small enough to sit inside the access point; we have not deployed it there. Edge-confined inference is the design target, not a measured deployment — but it is why we kept pushing on

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└What These Results Mean



- Let me be explicit about the boundary between what we proved and what we hope.
- We have shown that a self-supervised world model can classify four passenger movement states from commodity RSSI, above 0.92 MCC, with an edge-feasible compact variant, and that it stays stable when the collection grows while supervised baselines do not.
- We have not shown that this works on a moving vehicle. Our environment is two controlled indoor zones. A real bus adds vibration, metal reflections, crowding and body absorption, all of which change the RSSI distribution.
- The device pool is seven phone models, and we recorded no participant demographics, so we cannot claim equivalence across passenger groups.
- Most important, and stated in the paper: the training corpus overlaps the evaluation splits. The absolute values are

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└Conclusions and Future Work

Conclusions and Future Work

This work:

- **Self-supervised world models** for passive ISAC in transit
- Four JEPA variants, 18-channel hierarchy, calibrated ensembles
- **0.9194** single · **0.9232** ensemble · **0.9160** at 3.42 M
- **Stability under collection expansion**, where supervised models collapse

Models and code: github.com/ATNoG/inMotion

Next steps:

- **Cooperative multi-AP fusion** with spatial diversity
- **Online continual adaptation** to seasonal radio drift
- **Elastic privacy**: throttle resolution by motion state
- **Operational validation** on a vehicle in service
- Close the gap between single models (0.9194) and the 0.9563 ensemble result on the multi-device subset

- To conclude. We reformulated passenger movement classification as learning a world model of the radio channel, and it beats the supervised state of the art on this task by a wide margin: 0.9194 for a single model, 0.9232 for a calibrated ensemble, 0.9160 for a three-member subset of 3.42 million parameters.
- The result I would like you to remember is not the peak number but the direction of the arrows: as the collection grows, the self-supervised models improve and the supervised ones break. That is what a predictive objective over the signal buys you.
- The limitations are the agenda. We need multi-AP spatial diversity to resolve overlapping zones. We need online adaptation for seasonal drift. We need a real vehicle, in service, with energy and thermal measurements.
- Elastic privacy is the part I am most excited about: sensing

Acknowledgements

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Questions?

Code and models: github.com/ATNoG/inMotion · Paper: FNWF 2026

- Before questions, the funding acknowledgement: this work is part of inMotion, supported by European structural funds through CENTRO 2030.
- The models, the training scripts and the evaluation harness are public on GitHub, so every number in this talk can be reproduced.
- I am happy to take questions.